

mysql> show databases;

```
+-----+
| Database      |
+-----+
| information_schema |
| mysql         |
| nit           |
| performance_schema |
| sakila        |
| sys           |
| world         |
+-----+
```

7 rows in set (0.020 sec)

mysql> use nit;

Database changed

mysql> DROP TABLE IF EXISTS student;

Query OK, 0 rows affected (0.167 sec)

mysql> create table student(

-> name varchar(30),

-> id int not null primary key,

-> address varchar(50),

-> marks int);

Query OK, 0 rows affected (0.184 sec)

mysql> desc student;

Field	Type	Null	Key	Default	Extra
name	varchar(30)	YES		NULL	
id	int	NO	PRI	NULL	
address	varchar(50)	YES		NULL	
marks	int	YES		NULL	

4 rows in set (0.023 sec)

mysql> select * from student;

Empty set (0.012 sec)

mysql> insert into student(marks, id, name, address)
values(78, 12, 'Alfiya', 'hyd');

Query OK, 1 row affected (0.040 sec)

```
mysql> insert into student values('Khan', 40, 'bng', 66);
```

Query OK, 1 row affected (0.032 sec)

```
mysql> insert into student values('Alex', 45, 'hyd', 79),  
('Cathy', 17, 'Delhi', 90),
```

```
-> ('Dolly', 48, 'Pune', 67), ('Chancy', 78, 'Mumbai', 34);
```

Query OK, 4 rows affected (0.041 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> select * from student;
```

```
+-----+-----+-----+-----+  
| name  | id  | address | marks |  
+-----+-----+-----+-----+  
| Alfiya | 12 | hyd    | 78 |  
| Cathy  | 17 | Delhi  | 90 |  
| Khan   | 40 | bng    | 66 |  
| Alex   | 45 | hyd    | 79 |  
| Dolly  | 48 | Pune   | 67 |  
| Chancy | 78 | Mumbai | 34 |  
+-----+-----+-----+-----+
```

6 rows in set (0.005 sec)

```
mysql> select name from student;
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| Alfiya |
```

```
| Cathy |
```

```
| Khan |
```

```
| Alex |
```

```
| Dolly |
```

```
| Chancy |
```

```
+-----+
```

```
6 rows in set (0.005 sec)
```

```
mysql> select name, id from student;
```

```
+-----+-----+
```

```
| name | id |
```

```
+-----+-----+
```

```
| Alfiya | 12 |
```

```
| Cathy | 17 |
```

```
| Khan | 40 |
```

```
| Alex | 45 |
```

```
| Dolly | 48 |
```

| Chancy | 78 |

+-----+-----+

6 rows in set (0.005 sec)

mysql> select * from student where id = 12;

+-----+-----+-----+-----+

| name | id | address | marks |

+-----+-----+-----+-----+

| Alfiya | 12 | hyd | 78 |

+-----+-----+-----+-----+

1 row in set (0.006 sec)

mysql> select * from student;

+-----+-----+-----+-----+

| name | id | address | marks |

+-----+-----+-----+-----+

| Alfiya | 12 | hyd | 78 |

| Cathy | 17 | Delhi | 90 |

| Khan | 40 | bng | 66 |

| Alex | 45 | hyd | 79 |

| Dolly | 48 | Pune | 67 |

| Chancy | 78 | Mumbai | 34 |

+-----+-----+-----+-----+

6 rows in set (0.005 sec)

mysql> insert into student values('Sam', 12, 'hyd', 56);

**ERROR 1062 (23000): Duplicate entry '12' for key
'student.PRIMARY'**

**mysql> update student set address = "Chennai" Where id =
45;**

Query OK, 1 row affected (0.065 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from student;

+-----+-----+-----+-----+

| name | id | address | marks |

+-----+-----+-----+-----+

| Alfiya | 12 | hyd | 78 |

| Cathy | 17 | Delhi | 90 |

| Khan | 40 | bng | 66 |

| Alex | 45 | Chennai | 79 |

| Dolly | 48 | Pune | 67 |

| Chancy | 78 | Mumbai | 34 |

+-----+-----+-----+-----+

6 rows in set (0.005 sec)

mysql> alter table student add phoneNo int;

Query OK, 0 rows affected (0.151 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from student;

name	id	address	marks	phoneNo
Alfiya	12	hyd	78	NULL
Cathy	17	Delhi	90	NULL
Khan	40	bng	66	NULL
Alex	45	Chennai	79	NULL
Dolly	48	Pune	67	NULL
Chancy	78	Mumbai	34	NULL

6 rows in set (0.005 sec)

mysql> update student set phoneNo = 123;

Query OK, 6 rows affected (0.053 sec)

Rows matched: 6 Changed: 6 Warnings: 0

mysql> select * from student;

name	id	address	marks	phoneNo
Alfiya	12	hyd	78	123
Cathy	17	Delhi	90	123
Khan	40	bng	66	123
Alex	45	Chennai	79	123
Dolly	48	Pune	67	123
Chancy	78	Mumbai	34	123

6 rows in set (0.005 sec)

mysql> update student set phoneNo = 456 Where id = 12;

Query OK, 1 row affected (0.048 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from student;

name	id	address	marks	phoneNo
Alfiya	12	hyd	78	456

Cathy	17	Delhi	90	123
Khan	40	bng	66	123
Alex	45	Chennai	79	123
Dolly	48	Pune	67	123
Chancy	78	Mumbai	34	123

+-----+-----+-----+-----+-----+

6 rows in set (0.005 sec)

mysql> desc student;

Field	Type	Null	Key	Default	Extra
name	varchar(30)	YES		NULL	
id	int	NO	PRI	NULL	
address	varchar(50)	YES		NULL	
marks	int	YES		NULL	
phoneNo	int	YES		NULL	

5 rows in set (0.018 sec)

mysql> alter table student modify column name
varchar(60);

Query OK, 0 rows affected (0.103 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student;

Field	Type	Null	Key	Default	Extra
name	varchar(60)	YES		NULL	
id	int	NO	PRI	NULL	
address	varchar(50)	YES		NULL	
marks	int	YES		NULL	
phoneNo	int	YES		NULL	

5 rows in set (0.019 sec)

mysql> alter table student drop column phoneNo;

Query OK, 0 rows affected (0.172 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from student;

name	id	address	marks
------	----	---------	-------

```

+-----+----+-----+-----+
| Alfiya | 12 | hyd   | 78 |
| Cathy  | 17 | Delhi | 90 |
| Khan   | 40 | bng   | 66 |
| Alex   | 45 | Chennai | 79 |
| Dolly  | 48 | Pune  | 67 |
| Chancy | 78 | Mumbai | 34 |
+-----+----+-----+-----+

```

6 rows in set (0.006 sec)

mysql> delete from student where name = 'khan';

Query OK, 1 row affected (0.043 sec)

mysql> select * from student;

```

+-----+----+-----+-----+
| name  | id | address | marks |
+-----+----+-----+-----+
| Alfiya | 12 | hyd     | 78 |
| Cathy  | 17 | Delhi   | 90 |
| Alex   | 45 | Chennai | 79 |
| Dolly  | 48 | Pune    | 67 |
| Chancy | 78 | Mumbai  | 34 |

```

```
+-----+----+-----+-----+
```

5 rows in set (0.005 sec)

mysql> select sum(marks) from student;

```
+-----+
```

```
| sum(marks) |
```

```
+-----+
```

```
|    348 |
```

```
+-----+
```

1 row in set (0.007 sec)

mysql> select avg(marks) from student;

```
+-----+
```

```
| avg(marks) |
```

```
+-----+
```

```
|  69.6000 |
```

```
+-----+
```

1 row in set (0.006 sec)

mysql> select count(name) from student;

```
+-----+
```

```
| count(name) |
```

+-----+

| 5 |

+-----+

1 row in set (0.006 sec)

mysql> select max(marks) from student;

+-----+

| max(marks) |

+-----+

| 90 |

+-----+

1 row in set (0.006 sec)

mysql> select min(marks) from student;

+-----+

| min(marks) |

+-----+

| 34 |

+-----+

1 row in set (0.005 sec)

mysql> select * from student order by marks;

```

+-----+-----+-----+-----+
| name  | id  | address | marks |
+-----+-----+-----+-----+
| Chancy | 78 | Mumbai | 34 |
| Dolly  | 48 | Pune   | 67 |
| Alfiya | 12 | hyd    | 78 |
| Alex   | 45 | Chennai | 79 |
| Cathy  | 17 | Delhi   | 90 |
+-----+-----+-----+-----+
5 rows in set (0.006 sec)

```

mysql> select * from student order by marks desc;

```

+-----+-----+-----+-----+
| name  | id  | address | marks |
+-----+-----+-----+-----+
| Cathy | 17 | Delhi   | 90 |
| Alex  | 45 | Chennai | 79 |
| Alfiya | 12 | hyd    | 78 |
| Dolly  | 48 | Pune   | 67 |
| Chancy | 78 | Mumbai | 34 |
+-----+-----+-----+-----+
5 rows in set (0.006 sec)

```

mysql> select * from student where name like 'a%';

```
+-----+----+-----+-----+
| name  | id  | address | marks |
+-----+----+-----+-----+
| Alfiya | 12  | hyd     | 78    |
| Alex   | 45  | Chennai | 79    |
+-----+----+-----+-----+
```

2 rows in set (0.006 sec)

mysql> select * from student where name like '%y';

```
+-----+----+-----+-----+
| name  | id  | address | marks |
+-----+----+-----+-----+
| Cathy | 17  | Delhi   | 90    |
| Dolly | 48  | Pune    | 67    |
| Chancy | 78  | Mumbai  | 34    |
+-----+----+-----+-----+
```

3 rows in set (0.005 sec)

mysql> select * from student where name like '_a%';

```
+-----+----+-----+-----+
```

| name | id | address | marks |

+-----+----+-----+-----+

| Cathy | 17 | Delhi | 90 |

+-----+----+-----+-----+

1 row in set (0.007 sec)

mysql> select * from student where name like '%s_';

Empty set (0.006 sec)

mysql> DROP TABLE IF EXISTS emp;

Query OK, 0 rows affected (0.158 sec)

**mysql> create table emp(id int not null primary key, salary
int, empcode int, name varchar(30));**

Query OK, 0 rows affected (0.171 sec)

**mysql> insert into emp values(12, 20000, 102, 'Ayra'), (70,
30000, 108, 'Afjal'),**

-> (80, 25000, 103, 'Iram'), (30, 90000, 106, 'Sahil');

Query OK, 4 rows affected (0.039 sec)

Records: 4 Duplicates: 0 Warnings: 0

mysql> select * from emp;

+----+-----+-----+-----+

id	salary	empcode	name
12	20000	102	Ayra
30	90000	106	Sahil
70	30000	108	Afjal
80	25000	103	Iram

4 rows in set (0.005 sec)

```
mysql> select * from student;
```

name	id	address	marks
Alfiya	12	hyd	78
Cathy	17	Delhi	90
Alex	45	Chennai	79
Dolly	48	Pune	67
Chancy	78	Mumbai	34

5 rows in set (0.005 sec)

```
mysql> select * from student inner join emp on student. id
= emp.id;
```

```
+-----+----+-----+-----+----+-----+-----+-----+
| name  | id | address | marks | id | salary | empcode |
name |
+-----+----+-----+-----+----+-----+-----+-----+
| Alfiya | 12 | hyd    | 78 | 12 | 20000 | 102 | Ayra |
+-----+----+-----+-----+----+-----+-----+-----+

1 row in set (0.006 sec)
```

```
mysql> select * from student;
```

```
+-----+----+-----+-----+
| name  | id | address | marks |
+-----+----+-----+-----+
| Alfiya | 12 | hyd    | 78 |
| Cathy  | 17 | Delhi  | 90 |
| Alex   | 45 | Chennai | 79 |
| Dolly  | 48 | Pune   | 67 |
| Chancy | 78 | Mumbai | 34 |
+-----+----+-----+-----+

5 rows in set (0.005 sec)
```

```
mysql> select * from emp;
```

```

+----+-----+-----+-----+
| id | salary | empcode | name |
+----+-----+-----+-----+
| 12 | 20000 | 102 | Ayra |
| 30 | 90000 | 106 | Sahil |
| 70 | 30000 | 108 | Afjal |
| 80 | 25000 | 103 | Iram |
+----+-----+-----+-----+

```

4 rows in set (0.007 sec)

mysql> select * from student left join emp on student. id = emp.id;

```

+-----+----+-----+-----+-----+-----+-----+-----+
| name | id | address | marks | id | salary | empcode | name |
+-----+----+-----+-----+-----+-----+-----+-----+
| Alfiya | 12 | hyd | 78 | 12 | 20000 | 102 | Ayra |
| Cathy | 17 | Delhi | 90 | NULL | NULL | NULL | NULL |
| Alex | 45 | Chennai | 79 | NULL | NULL | NULL | NULL |
| Dolly | 48 | Pune | 67 | NULL | NULL | NULL | NULL |

```

Chancy 78 Mumbai 34 NULL NULL NULL NULL

+-----+-----+-----+-----+-----+-----+-----+-----+

5 rows in set (0.006 sec)

```
mysql> select * from student right join emp on student. id = emp.id;
```

+-----+-----+-----+-----+-----+-----+-----+-----+

name id address marks id salary empcode
name

+-----+-----+-----+-----+-----+-----+-----+-----+

Alfiya 12 hyd 78 12 20000 102 Ayra
--

NULL NULL NULL NULL 30 90000 106
Sahil

NULL NULL NULL NULL 70 30000 108
Afjal

NULL NULL NULL NULL 80 25000 103
Iram

+-----+-----+-----+-----+-----+-----+-----+-----+

4 rows in set (0.006 sec)

```
mysql> select * from student cross join emp;
```

+-----+-----+-----+-----+-----+-----+-----+-----+

**| name | id | address | marks | id | salary | empcode |
name |**

+-----+----+-----+-----+----+-----+-----+-----+

| Alfiya | 12 | hyd | 78 | 80 | 25000 | 103 | Iram |

| Alfiya | 12 | hyd | 78 | 70 | 30000 | 108 | Afjal |

| Alfiya | 12 | hyd | 78 | 30 | 90000 | 106 | Sahil |

| Alfiya | 12 | hyd | 78 | 12 | 20000 | 102 | Ayra |

| Cathy | 17 | Delhi | 90 | 80 | 25000 | 103 | Iram |

| Cathy | 17 | Delhi | 90 | 70 | 30000 | 108 | Afjal |

| Cathy | 17 | Delhi | 90 | 30 | 90000 | 106 | Sahil |

| Cathy | 17 | Delhi | 90 | 12 | 20000 | 102 | Ayra |

| Alex | 45 | Chennai | 79 | 80 | 25000 | 103 | Iram |

| Alex | 45 | Chennai | 79 | 70 | 30000 | 108 | Afjal |

| Alex | 45 | Chennai | 79 | 30 | 90000 | 106 | Sahil |

| Alex | 45 | Chennai | 79 | 12 | 20000 | 102 | Ayra |

| Dolly | 48 | Pune | 67 | 80 | 25000 | 103 | Iram |

| Dolly | 48 | Pune | 67 | 70 | 30000 | 108 | Afjal |

| Dolly | 48 | Pune | 67 | 30 | 90000 | 106 | Sahil |

| Dolly | 48 | Pune | 67 | 12 | 20000 | 102 | Ayra |

| Chancy | 78 | Mumbai | 34 | 80 | 25000 | 103 | Iram |

| Chancy | 78 | Mumbai | 34 | 70 | 30000 | 108 | Afjal |

| Chancy | 78 | Mumbai | 34 | 30 | 90000 | 106 | Sahil |

Chancy	78	Mumbai	34	12	20000	102	Ayra
--------	----	--------	----	----	-------	-----	------

20 rows in set (0.006 sec)

```
mysql> select * from student inner join emp on student.id
= emp.id;
```

name	id	address	marks	id	salary	empcode	name
------	----	---------	-------	----	--------	---------	------

Alfiya	12	hyd	78	12	20000	102	Ayra
--------	----	-----	----	----	-------	-----	------

1 row in set (0.006 sec)

```
mysql> show tables;
```

Tables_in_nit
emp
student
student1
student2

4 rows in set (0.018 sec)

mysql> DROP TABLE IF EXISTS student1;

Query OK, 0 rows affected (0.150 sec)

mysql> create table student1(

-> name varchar(30),

-> id INT);

Query OK, 0 rows affected (0.163 sec)

**mysql> insert into student1 values ('bin', 12), ('bin', 12),
('bin', 12),**

-> ('kim', 30), ('Hari', 45), ('Rahul', 40);

Query OK, 6 rows affected (0.041 sec)

Records: 6 Duplicates: 0 Warnings: 0

mysql> start transaction;

Query OK, 0 rows affected (0.004 sec)

mysql> insert into student1 values('bin1', 70);

Query OK, 1 row affected (0.007 sec)

mysql> select * from student1;

```
+-----+-----+
```

```
| name | id |
```

```
+-----+-----+
```

```
| bin  | 12 |
```

```
| bin  | 12 |
```

```
| bin  | 12 |
```

```
| kim  | 30 |
```

```
| Hari | 45 |
```

```
| Rahul | 40 |
```

```
| bin1  | 70 |
```

```
+-----+-----+
```

7 rows in set (0.006 sec)

mysql> delete from student1 where name = 'Hari';

Query OK, 1 row affected (0.007 sec)

mysql> select * from student1;

```
+-----+-----+
```

```
| name | id |
```

```
+-----+-----+
```

```
| bin  | 12 |
```

```
| bin  | 12 |
```


bin	12
kim	30
Rahul	40
bin1	70

+-----+-----+

6 rows in set (0.005 sec)

mysql> rollback;

Query OK, 0 rows affected (0.019 sec)

mysql> select * from student1;

+-----+-----+

name	id
------	----

+-----+-----+

bin	12
-----	----

bin	12
-----	----

bin	12
-----	----

kim	30
-----	----

Hari	45
------	----

Rahul	40
-------	----

+-----+-----+

6 rows in set (0.005 sec)

mysql> insert into student1 values('welcome', 20);

Query OK, 1 row affected (0.033 sec)

mysql> select * from student1;

+-----+-----+

| name | id |

+-----+-----+

| bin | 12 |

| bin | 12 |

| bin | 12 |

| kim | 30 |

| Hari | 45 |

| Rahul | 40 |

| welcome | 20 |

+-----+-----+

7 rows in set (0.005 sec)

mysql> commit;

Query OK, 0 rows affected (0.003 sec)

mysql> rollback;

Query OK, 0 rows affected (0.004 sec)

mysql> select * from student1;

+-----+-----+

| name | id |

+-----+-----+

| bin | 12 |

| bin | 12 |

| bin | 12 |

| kim | 30 |

| Hari | 45 |

| Rahul | 40 |

| welcome | 20 |

+-----+-----+

7 rows in set (0.005 sec)

mysql> rollback;

Query OK, 0 rows affected (0.004 sec)

mysql> select * from student1;

+-----+-----+

| name | id |

+-----+-----+

| bin | 12 |

| bin | 12 |

| bin | 12 |

| kim | 30 |

| Hari | 45 |

| Rahul | 40 |

| welcome | 20 |

+-----+-----+

7 rows in set (0.005 sec)

mysql> start transaction;

Query OK, 0 rows affected (0.005 sec)

mysql> select * from student1;

+-----+-----+

| name | id |

+-----+-----+

| bin | 12 |

| bin | 12 |

| bin | 12 |

| kim | 30 |

Hari	45	
Rahul	40	
welcome	20	

+-----+-----+

7 rows in set (0.007 sec)

mysql> delete from student1 where name = 'Rahul';

Query OK, 1 row affected (0.008 sec)

mysql> select * from student1;

+-----+-----+

name	id	
------	----	--

+-----+-----+

bin	12	
-----	----	--

bin	12	
-----	----	--

bin	12	
-----	----	--

kim	30	
-----	----	--

Hari	45	
------	----	--

welcome	20	
---------	----	--

+-----+-----+

6 rows in set (0.005 sec)

mysql> rollback;

Query OK, 0 rows affected (0.014 sec)

mysql> select * from student1;

+-----+-----+

| name | id |

+-----+-----+

| bin | 12 |

| bin | 12 |

| bin | 12 |

| kim | 30 |

| Hari | 45 |

| Rahul | 40 |

| welcome | 20 |

+-----+-----+

7 rows in set (0.006 sec)

mysql> start transaction;

Query OK, 0 rows affected (0.003 sec)

mysql> insert into student1 values('Faisal', 45);

Query OK, 1 row affected (0.007 sec)

mysql> savepoint kamal;

Query OK, 0 rows affected (0.003 sec)

mysql> insert into student1 values('Amir', 85);

Query OK, 1 row affected (0.005 sec)

mysql> savepoint kamal;

Query OK, 0 rows affected (0.003 sec)

mysql> select * from student1;

+-----+-----+

| name | id |

+-----+-----+

| bin | 12 |

| bin | 12 |

| bin | 12 |

| kim | 30 |

| Hari | 45 |

| Rahul | 40 |

| welcome | 20 |

| Faisal | 45 |

Amir	85	
------	----	--

+-----+-----+

9 rows in set (0.006 sec)

mysql> rollback to kamal;

Query OK, 0 rows affected (0.003 sec)

mysql> select * from student1;

+-----+-----+

name	id	
------	----	--

+-----+-----+

bin	12	
-----	----	--

bin	12	
-----	----	--

bin	12	
-----	----	--

kim	30	
-----	----	--

Hari	45	
------	----	--

Rahul	40	
-------	----	--

welcome	20	
---------	----	--

Faisal	45	
--------	----	--

Amir	85	
------	----	--

+-----+-----+

9 rows in set (0.005 sec)

mysql> insert into student1 values('Arbaz', 95);

Query OK, 1 row affected (0.006 sec)

mysql> savepoint kamal;

Query OK, 0 rows affected (0.004 sec)

mysql> rollback to kamal;

Query OK, 0 rows affected (0.003 sec)

mysql> select * from student1;

+-----+-----+

| name | id |

+-----+-----+

| bin | 12 |

| bin | 12 |

| bin | 12 |

| kim | 30 |

| Hari | 45 |

| Rahul | 40 |

| welcome | 20 |

| Faisal | 45 |

Amir	85
------	----

Arbaz	95
-------	----

+-----+-----+

10 rows in set (0.005 sec)

mysql> delete from student1 where name = 'Arbaz';

Query OK, 1 row affected (0.009 sec)

mysql> select * from student1;

+-----+-----+

name	id
------	----

+-----+-----+

bin	12
-----	----

bin	12
-----	----

bin	12
-----	----

kim	30
-----	----

Hari	45
------	----

Rahul	40
-------	----

welcome	20
---------	----

Faisal	45
--------	----

Amir	85
------	----

+-----+-----+

9 rows in set (0.005 sec)

mysql> rollback to kamal;

Query OK, 0 rows affected (0.004 sec)

mysql> select * from student1;

```
+-----+-----+
| name  | id  |
+-----+-----+
| bin   | 12  |
| bin   | 12  |
| bin   | 12  |
| kim   | 30  |
| Hari  | 45  |
| Rahul | 40  |
| welcome | 20 |
| Faisal | 45 |
| Amir  | 85 |
| Arbaz  | 95 |
+-----+-----+
```

10 rows in set (0.006 sec)

mysql> create database ddlcommand;

Query OK, 1 row affected (0.105 sec)

mysql> show databases;

```
+-----+
| Database      |
+-----+
| ddlcommand    |
| information_schema |
| mysql          |
| nit           |
| performance_schema |
| sakila         |
| sys           |
| world         |
+-----+
```

8 rows in set (0.012 sec)

mysql> DROP TABLE IF EXISTS student;

Query OK, 0 rows affected (0.188 sec)

**mysql> create table student(name varchar(30), roll_no int
not null primary key, marks int);**

Query OK, 0 rows affected (0.165 sec)

mysql> desc student;

Field	Type	Null	Key	Default	Extra
name	varchar(30)	YES		NULL	
roll_no	int	NO	PRI	NULL	
marks	int	YES		NULL	

3 rows in set (0.024 sec)

mysql> alter table student add addr varchar(50);

Query OK, 0 rows affected (0.616 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student;

Field	Type	Null	Key	Default	Extra
name	varchar(30)	YES		NULL	
roll_no	int	NO	PRI	NULL	
marks	int	YES		NULL	
addr	varchar(50)	YES		NULL	

Field	Type	Null	Key	Default	Extra
addr	varchar(50)	YES		NULL	

4 rows in set (0.017 sec)

mysql> alter table student drop marks;

Query OK, 0 rows affected (0.412 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student;

Field	Type	Null	Key	Default	Extra
name	varchar(30)	YES		NULL	
roll_no	int	NO	PRI	NULL	
addr	varchar(50)	YES		NULL	

3 rows in set (0.021 sec)

mysql> alter table student modify name varchar(50);

Query OK, 0 rows affected (0.092 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student;

Field	Type	Null	Key	Default	Extra
name	varchar(50)	YES		NULL	
roll_no	int	NO	PRI	NULL	
addr	varchar(50)	YES		NULL	

3 rows in set (0.018 sec)

mysql> alter table student drop primary key;

Query OK, 0 rows affected (0.450 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student;

Field	Type	Null	Key	Default	Extra
name	varchar(50)	YES		NULL	
roll_no	int	NO		NULL	
addr	varchar(50)	YES		NULL	

3 rows in set (0.018 sec)

mysql> alter table student add primary key(name);

Query OK, 0 rows affected (0.349 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student;

Field	Type	Null	Key	Default	Extra
name	varchar(50)	NO	PRI	NULL	
roll_no	int	NO		NULL	
addr	varchar(50)	YES		NULL	

3 rows in set (0.017 sec)

mysql> rename table student to std;

Query OK, 0 rows affected (0.174 sec)

mysql> desc std;

Field	Type	Null	Key	Default	Extra
name	varchar(50)	NO	PRI	NULL	

roll_no	int	NO	NULL	
addr	varchar(50)	YES	NULL	
+-----+-----+-----+-----+-----+				

3 rows in set (0.024 sec)

mysql> insert into std values('nit', 6, 'Amritsar');

Query OK, 1 row affected (0.054 sec)

mysql> select * from std;

+-----+-----+-----+		
name	roll_no	addr
+-----+-----+-----+		
nit	6	Amritsar
+-----+-----+-----+		

1 row in set (0.005 sec)

mysql> truncate table std;

Query OK, 0 rows affected (0.251 sec)

mysql> select * from std;

Empty set (0.014 sec)

mysql> desc std;

Field	Type	Null	Key	Default	Extra
name	varchar(50)	NO	PRI	NULL	
roll_no	int	NO		NULL	
addr	varchar(50)	YES		NULL	

3 rows in set (0.019 sec)

mysql> drop table std;

Query OK, 0 rows affected (0.147 sec)

mysql> CREATE DATABASE primarykey;

Query OK, 1 row affected (0.065 sec)

mysql> use primarykey;

Database changed

mysql> DROP TABLE IF EXISTS student;

Query OK, 0 rows affected, 1 warning (0.028 sec)

**mysql> create table student(id int not null primary key,
name varchar(30), age int);**

Query OK, 0 rows affected (0.198 sec)

mysql> desc student;

+-----+-----+-----+-----+-----+-----+					
Field	Type	Null	Key	Default	Extra
+-----+-----+-----+-----+-----+-----+					
id	int	NO	PRI	NULL	
name	varchar(30)	YES		NULL	
age	int	YES		NULL	

+-----+-----+-----+-----+-----+-----+

3 rows in set (0.023 sec)

mysql> DROP TABLE IF EXISTS student1;

Query OK, 0 rows affected, 1 warning (0.030 sec)

mysql> create table student1(id int not null, name varchar(50), addr varchar(50), primary key(id));

Query OK, 0 rows affected (0.223 sec)

mysql> desc student1;

+-----+-----+-----+-----+-----+-----+					
Field	Type	Null	Key	Default	Extra
+-----+-----+-----+-----+-----+-----+					

id	int	NO	PRI	NULL		
name	varchar(50)	YES		NULL		
addr	varchar(50)	YES		NULL		
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+

3 rows in set (0.023 sec)

mysql> DROP TABLE IF EXISTS student2;

Query OK, 0 rows affected, 1 warning (0.030 sec)

mysql> CREATE TABLE student2 (id INT NOT NULL, name VARCHAR(50), age INT, CONSTRAINT pk_student PRIMARY KEY (id, name));

Query OK, 0 rows affected (0.215 sec)

mysql> desc student2;

+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+
Field	Type	Null	Key	Default	Extra	
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+
id	int	NO	PRI	NULL		
name	varchar(50)	NO	PRI	NULL		
age	int	YES		NULL		
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+

3 rows in set (0.025 sec)

mysql> DROP TABLE IF EXISTS emp;

Query OK, 0 rows affected, 1 warning (0.035 sec)

**mysql> create table emp(id int, name varchar(40), addr
varchar(50));**

Query OK, 0 rows affected (0.177 sec)

mysql> desc emp;

Field	Type	Null	Key	Default	Extra
id	int	YES		NULL	
name	varchar(40)	YES		NULL	
addr	varchar(50)	YES		NULL	

3 rows in set (0.022 sec)

mysql> alter table emp add primary key(id);

Query OK, 0 rows affected (0.368 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc emp;

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	
name	varchar(40)	YES		NULL	
addr	varchar(50)	YES		NULL	

3 rows in set (0.017 sec)

mysql> DROP TABLE IF EXISTS emp;

Query OK, 0 rows affected (0.165 sec)

mysql> create table emp(id int, name varchar(40), age int);

Query OK, 0 rows affected (0.206 sec)

mysql> desc emp;

Field	Type	Null	Key	Default	Extra
id	int	YES		NULL	
name	varchar(40)	YES		NULL	
age	int	YES		NULL	

+-----+-----+-----+-----+-----+-----+

3 rows in set (0.035 sec)

mysql> create table em(id int, name varchar(40), age int);

Query OK, 0 rows affected (0.179 sec)

mysql> desc emp;

+-----+-----+-----+-----+-----+-----+

| Field | Type | Null | Key | Default | Extra |

+-----+-----+-----+-----+-----+-----+

| id | int | YES | | NULL | |

| name | varchar(40) | YES | | NULL | |

| age | int | YES | | NULL | |

+-----+-----+-----+-----+-----+-----+

3 rows in set (0.017 sec)

mysql> desc em;

+-----+-----+-----+-----+-----+-----+

| Field | Type | Null | Key | Default | Extra |

+-----+-----+-----+-----+-----+-----+

| id | int | YES | | NULL | |

| name | varchar(40) | YES | | NULL | |

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	
name	varchar(40)	YES		NULL	
age	int	NO	PRI	NULL	

3 rows in set (0.022 sec)

mysql> alter table em add constraint em primary key(id, age);

Query OK, 0 rows affected (0.359 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc em;

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	
name	varchar(40)	YES		NULL	
age	int	NO	PRI	NULL	

3 rows in set (0.018 sec)

mysql> alter table em drop primary key;

Query OK, 0 rows affected (0.418 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc em;

Field	Type	Null	Key	Default	Extra
id	int	NO		NULL	
name	varchar(40)	YES		NULL	
age	int	NO		NULL	

3 rows in set (0.018 sec)

mysql> DROP TABLE IF EXISTS employee;

Query OK, 0 rows affected, 1 warning (0.025 sec)

mysql> create database employee;

Query OK, 1 row affected (0.076 sec)

mysql> use employee;

Database changed

**mysql> create table person(personID int not null primary
key,**

-> name varchar(30), age int);

Query OK, 0 rows affected (0.179 sec)

mysql> desc person;

Field	Type	Null	Key	Default	Extra
personID	int	NO	PRI	NULL	
name	varchar(30)	YES		NULL	
age	int	YES		NULL	

3 rows in set (0.020 sec)

**mysql> create table personalorder(orderID int not null
primary key, ordernumber int, personID int,**

-> foreign key(personID) references person(personID));

Query OK, 0 rows affected (0.336 sec)

mysql> desc personalorder;

Field	Type	Null	Key	Default	Extra
orderID	int	NO	PRI	NULL	
ordernumber	int	YES		NULL	

personID	int	YES	MUL	NULL		
----------	-----	-----	-----	------	--	--

3 rows in set (0.022 sec)

```
mysql> create table personorder(orderID int,
ordernumber int, orderpersion int);
```

Query OK, 0 rows affected (0.192 sec)

```
mysql> desc personorder;
```

Field	Type	Null	Key	Default	Extra
orderID	int	YES		NULL	
ordernumber	int	YES		NULL	
orderpersion	int	YES		NULL	

3 rows in set (0.027 sec)

```
mysql> alter table personorder add foreign
key(orderpersion)
```

```
-> references person(personID);
```

Query OK, 0 rows affected (0.748 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc personorder;

Field	Type	Null	Key	Default	Extra
orderID	int	YES		NULL	
ordernumber	int	YES		NULL	
orderpersion	int	YES	MUL	NULL	

3 rows in set (0.016 sec)

mysql> alter table personorder add primary key(orderID);

Query OK, 0 rows affected (0.582 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> desc personorder;

Field	Type	Null	Key	Default	Extra
orderID	int	NO	PRI	NULL	
ordernumber	int	YES		NULL	
orderpersion	int	YES	MUL	NULL	

+-----+-----+-----+-----+-----+-----+

3 rows in set (0.025 sec)

mysql> DROP TABLE IF EXISTS student1;

Query OK, 0 rows affected (0.157 sec)

**mysql> create table student1(name varchar(30), id int
primary key, branch varchar(40));**

Query OK, 0 rows affected (0.170 sec)

mysql> DROP TABLE IF EXISTS student2;

Query OK, 0 rows affected (0.147 sec)

**mysql> create table student2(rollno int primary key, addr
varchar(50), phone int);**

Query OK, 0 rows affected (0.203 sec)

mysql> show tables;

+-----+

| Tables_in_employee |

+-----+

| person |

| personalorder |

| personorder |

```
| student1      |
| student2      |
+-----+
```

5 rows in set (0.017 sec)

```
mysql> insert into student1 values('John', 23, 'IT'), ('Alex',
35, 'CS'), ('Sana', 13, 'CS'), ('Rayan' , 25, 'Me');
```

Query OK, 4 rows affected (0.052 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> insert into student2 values(45, 'Delhi', 234), (35,
'Mumbai', 677),(70, 'Jaipur', 987), (25, 'Nagpur', 208);
```

Query OK, 4 rows affected (0.055 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> select * from student1;
```

```
+-----+-----+-----+
| name  | id   | branch |
+-----+-----+-----+
| Sana  | 13   | CS     |
| John  | 23   | IT     |
| Rayan | 25   | Me     |
| Alex  | 35   | CS     |
```

+-----+-----+-----+

4 rows in set (0.006 sec)

mysql> select * from student2;

+-----+-----+-----+

| rollno | addr | phone |

+-----+-----+-----+

| 25 | Nagpur | 208 |

| 35 | Mumbai | 677 |

| 45 | Delhi | 234 |

| 70 | Jaipur | 987 |

+-----+-----+-----+

4 rows in set (0.005 sec)

**mysql> select * from student1 inner join student2 on
student1.id = student2.rollno;**

+-----+-----+-----+-----+-----+-----+

| name | id | branch | rollno | addr | phone |

+-----+-----+-----+-----+-----+-----+

| Rayan | 25 | Me | 25 | Nagpur | 208 |

| Alex | 35 | CS | 35 | Mumbai | 677 |

+-----+-----+-----+-----+-----+-----+

2 rows in set (0.006 sec)

```
mysql> select name, branch, id from student1 inner join  
student2 on student1.id = student2.rollno;
```

```
+-----+-----+----+  
| name | branch | id |  
+-----+-----+----+  
| Rayan | Me    | 25 |  
| Alex  | CS    | 35 |  
+-----+-----+----+
```

2 rows in set (0.006 sec)

```
mysql> select student1.name, student2.rollno from  
student1 inner join student2 on student1.id =  
student2.rollno;
```

```
+-----+-----+  
| name | rollno |  
+-----+-----+  
| Rayan | 25 |  
| Alex  | 35 |  
+-----+-----+
```

2 rows in set (0.007 sec)


```
mysql> select * from student1 left join student2 on  
student1.id = student2.rollno;
```

```
+-----+-----+-----+-----+-----+-----+  
| name | id | branch | rollno | addr | phone |  
+-----+-----+-----+-----+-----+-----+  
| Sana | 13 | CS     | NULL | NULL | NULL |  
| John | 23 | IT     | NULL | NULL | NULL |  
| Rayan | 25 | Me     | 25 | Nagpur | 208 |  
| Alex | 35 | CS     | 35 | Mumbai | 677 |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.006 sec)
```

```
mysql> select * from student1 right join student2 on  
student1.id = student2.rollno;
```

```
+-----+-----+-----+-----+-----+-----+  
| name | id | branch | rollno | addr | phone |  
+-----+-----+-----+-----+-----+-----+  
| Rayan | 25 | Me     | 25 | Nagpur | 208 |  
| Alex | 35 | CS     | 35 | Mumbai | 677 |  
| NULL | NULL | NULL | 45 | Delhi | 234 |  
| NULL | NULL | NULL | 70 | Jaipur | 987 |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.006 sec)
```

```
mysql> select * from student1 cross join student2 on  
student1.id = student2.rollno;
```

```
+-----+-----+-----+-----+-----+-----+  
| name | id | branch | rollno | addr | phone |  
+-----+-----+-----+-----+-----+-----+  
| Rayan | 25 | Me | 25 | Nagpur | 208 |  
| Alex | 35 | CS | 35 | Mumbai | 677 |
```

```
+-----+-----+-----+-----+-----+-----+
```

2 rows in set (0.006 sec)

```
mysql> select * from student1 cross join student2;
```

```
+-----+-----+-----+-----+-----+-----+  
| name | id | branch | rollno | addr | phone |  
+-----+-----+-----+-----+-----+-----+  
| Alex | 35 | CS | 25 | Nagpur | 208 |  
| Rayan | 25 | Me | 25 | Nagpur | 208 |  
| John | 23 | IT | 25 | Nagpur | 208 |  
| Sana | 13 | CS | 25 | Nagpur | 208 |  
| Alex | 35 | CS | 35 | Mumbai | 677 |  
| Rayan | 25 | Me | 35 | Mumbai | 677 |  
| John | 23 | IT | 35 | Mumbai | 677 |
```

Sana 13 CS 35 Mumbai 677
Alex 35 CS 45 Delhi 234
Rayan 25 Me 45 Delhi 234
John 23 IT 45 Delhi 234
Sana 13 CS 45 Delhi 234
Alex 35 CS 70 Jaipur 987
Rayan 25 Me 70 Jaipur 987
John 23 IT 70 Jaipur 987
Sana 13 CS 70 Jaipur 987

+-----+-----+-----+-----+-----+-----+

16 rows in set (0.007 sec)

mysql> select user();

+-----+

user()

+-----+

root@localhost

+-----+

1 row in set (0.010 sec)

mysql> create user 'Alfiya'@localhost identified by 'Alfiya';

'> ^C

mysql> create user 'Alfiya'@'localhost' identified by 'Alfiya';

Query OK, 0 rows affected (0.236 sec)

mysql> show grants for 'Alfiya'@'localhost';

+-----+

| Grants for Alfiya@localhost |

+-----+

| GRANT USAGE ON *.* TO `Alfiya`@`localhost` |

+-----+

1 row in set (0.018 sec)

mysql> grant select on *.* to 'Alfiya'@'localhost';

Query OK, 0 rows affected (0.058 sec)

mysql> grant insert on employee.student1 to 'Alfiya'@'localhost';

Query OK, 0 rows affected (0.051 sec)

mysql>